Natural Hallucinations: Cross-Model Error Transfer in Large Language Models

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Abstract

Large language models hallucinate—but are all hallucinations created equal? We investigate whether certain factual errors are *natural hallucinations*: systematic failures that persist across models, resist prompt variation, and evade self-detection even when models are presented with correct answers. We evaluate four OpenAI models (GPT-4.1, GPT-4O, GPT-4O-MINI, and GPT-3.5-TURBO) on all 817 TRUTHFULQA questions and find that 27 questions (3.3%) cause every model to hallucinate, while 82 questions (10.0%) fool three or more models. These shared errors are robust: 76% of hallucinations persist across five question rephrasings. More strikingly, models detect only 10–16% of their own hallucinations in an A/B test against the correct answer—far below the 50% chance baseline and dramatically below their 66–95% accuracy on control questions. Cross-model overlap is 3–6× above random (Jaccard 0.19–0.41, all $p < 0.001$), and older-model errors predict newer-model errors ($r = 0.53$ for GPT-4O-MINI →GPT-4O). These findings suggest that a core set of hallucinations are structural features of how language models learn from data, not random errors that improved training will easily eliminate.

1 Introduction

Large language models produce factual errors. Despite rapid progress in scale, training data, and alignment techniques, hallucinations remain a persistent failure mode across model families and generations [Lin et al., 2022, Huang et al., 2023]. A common assumption is that these errors are essentially random—artifacts of insufficient training data or imperfect optimization that will gradually disappear as models improve. But what if some hallucinations are not random at all?

We investigate this question by studying what we call *natural hallucinations*: factual errors that multiple models consistently produce, that survive question rephrasing, and that models cannot recognize as wrong even when presented with the correct answer side by side. If such hallucinations exist, they represent a qualitatively different challenge from random errors—one that scaling alone is unlikely to solve.

Why this matters. The distinction between random and natural hallucinations has direct practical consequences. Random errors can be addressed through better data curation, larger models, or straightforward prompting strategies. Natural hallucinations, by contrast, may require fundamentally different interventions: targeted training data corrections, architectural changes, or external verification systems. Understanding which hallucinations are natural is therefore critical for both safety research and deployment decisions.

Gap in existing work. Prior work has established that LLMs produce "imitative falsehoods" that match common human misconceptions [Lin et al., 2022], that some hallucinations persist despite the model encoding correct information internally [Orgad et al., 2025, Simhi et al., 2025], and that training data frequency drives hallucination rates [Miao and Kearns, 2026]. However, no study has systematically characterized the *same set of questions* across all four dimensions: cross-model

transfer, robustness to rephrasing, failure of self-detection, and temporal prediction from older to newer models.

Our approach. We test four OpenAI models—GPT-4.1, GPT-4O, GPT-4O-MINI, and GPT-3.5-TURBO—on all 817 TRUTHFULQA questions. We identify questions that cause failures across multiple models, then probe these shared errors through three follow-up experiments: rephrasing robustness, self-detection with evidence, and cross-model transfer analysis. Our results paint a consistent picture: a core set of hallucinations behaves as a structural feature of language modeling, not as a bug to be patched.

Key findings. We find that 27 questions (3.3%) cause all four models to hallucinate, with 82 questions (10.0%) fooling three or more. These hallucinations persist at a 76% rate across five rephrasings per question. In an A/B self-detection test, models choose the correct answer only 10–16% of the time—worse than random—while achieving 66–95% on control questions where they originally answered correctly. Cross-model Jaccard similarity ranges from 0.19 to 0.41 (all $p < 0.001$), and GPT-4O-MINI hallucinations predict GPT-4O hallucinations with $r = 0.53$.

In summary, our main contributions are:

- We identify 27 universal and 82 near-universal hallucinations that persist across four models spanning three years of development, establishing the existence of natural hallucinations.
- We demonstrate that these errors are robust to rephrasing (76% persistence) and resistant to self-detection (10–16% accuracy, $z = 18.49$, $p < 0.0001$), revealing a systematic bias rather than a surface-level pattern match.
- We show that hallucination sets overlap 3–6$\times$ above random across model pairs and that older-model errors significantly predict newer-model errors ($r = 0.26$–0.53), suggesting shared training data biases as a root cause.

2 Related Work

Hallucination characterization and benchmarks. Lin et al. [2022] introduced TRUTHFULQA, a benchmark of 817 questions designed to elicit "imitative falsehoods"—incorrect answers that LLMs produce because they appear frequently in training data. They found an inverse scaling pattern where larger models were *less* truthful, suggesting that scale amplifies training data biases rather than correcting them. Li et al. [2023] developed HALUEVAL, a large-scale benchmark for hallucination detection across QA, dialogue, and summarization. Gekhman et al. [2024] conducted a systematic study of factual error patterns across models, finding consistent category-level trends. We build on TRUTHFULQA as our primary evaluation, but go beyond per-model accuracy to study the *transfer* and *persistence* of specific hallucinated instances across models.

Knowledge versus output: errors despite knowing. A growing body of work demonstrates that LLMs can encode correct information internally while still producing incorrect outputs. Orgad et al. [2025] showed that linear probes on intermediate representations can detect truthfulness with AUC 0.85–0.95, even for "Type C" errors where the model gives the same wrong answer 29 out of 30 times. Simhi et al. [2024] formalized the distinction between ignorance (HK$-$: the model lacks knowledge) and error (HK$+$: the model knows but still hallucinates), finding that 4–24% of hallucinations are HK$+$. Simhi et al. [2025] defined CHOKE (Certain Hallucinations Overriding Known Evidence) as cases where models hallucinate with high certainty despite demonstrably possessing the correct answer, finding that 16–43% of knowledge-despite-hallucination cases exhibit this pattern. Our self-detection experiment complements this line of work: rather than probing hidden states, we test whether models can *recognize* their own errors when the correct answer is explicitly presented alongside their hallucinated response.

Training data frequency and hallucination. Miao and Kearns [2026] provided theoretical grounding for the connection between training data frequency and hallucination, showing that calibrated LLMs must hallucinate at a rate lower-bounded by the "monofact" rate—the fraction of facts appearing only once in training data. Niimi [2025] found that hallucination rates are strongly correlated with citation counts (a proxy for training data frequency), with an inflection point at approximately 90 citations and saturation at 1,248. These results suggest that natural hallucinations may cluster around facts that contradict high-frequency misconceptions in training data—precisely the design principle behind TRUTHFULQA.

Cross-model consistency and transfer. McKenna et al. [2023] demonstrated that attestation bias—the tendency to affirm training-attested hypotheses regardless of context—appears consistently across LLaMA, GPT-3.5, PaLM, and GPT-4. Finch and Zk [2025] studied cross-model consistency as a signal for shared error patterns. However, Simhi et al. [2024] found relatively low cross-model Jaccard similarity (0.10–0.36) for HK+ instances specifically, suggesting that *which* questions trigger errors may vary even when the underlying phenomenon transfers. Our work directly measures cross-model overlap of hallucinated question sets using the complete TRUTHFULQA benchmark, finding higher overlap (0.19–0.41) than Simhi et al. [2024] and significant temporal prediction from older to newer models.

Theoretical perspectives on hallucination. Several recent works argue that hallucination is a fundamental rather than incidental property of language models. Xu et al. [2024] proved formally that LLMs with computable training procedures cannot avoid hallucination on all inputs. Kalai [2025] showed that natural statistical pressures inherent to language modeling cause systematic hallucination patterns. Banerjee et al. [2024] argued from a Gödelian perspective that structural hallucination is unavoidable. Hager et al. [2026] found that models encode uncertainty signals internally but these signals are often suppressed at the output layer. Our empirical findings are consistent with these theoretical accounts: the natural hallucinations we identify appear to be structural features of how language models process widely-believed but incorrect information.

Mitigation approaches. Dhuliawala et al. [2023] proposed chain-of-verification, which reduces hallucinations by having models verify their own outputs. Feng et al. [2024] explored multi-LLM collaboration for identifying knowledge gaps and encouraging abstention. Sharma et al. [2024] demonstrated that RLHF training can exacerbate sycophancy, a related failure mode. Our self-detection results (10–16% accuracy) suggest that verification-based approaches may be fundamentally limited for natural hallucinations, since the model's internal bias toward the wrong answer persists even under direct comparison with the truth.

3 Methodology

We design a four-experiment pipeline to test whether certain hallucinations are *natural*—consistent across models, robust to rephrasing, resistant to self-detection, and predictable across generations.

3.1 Dataset

We use the complete TRUTHFULQA benchmark [Lin et al., 2022], consisting of 817 questions across 38 categories including Misconceptions, Superstitions, Myths, Education, Misquotations, and Conspiracies. Each question includes a best answer, a set of correct answers, and a set of incorrect answers. The benchmark is designed to elicit "imitative falsehoods"—answers that sound plausible because they reflect widely-held but incorrect beliefs. We load the dataset directly from HuggingFace (`truthfulqa/truthful_qa`, generation configuration) without filtering or modification.

3.2 Models

We evaluate four OpenAI models spanning three years of development:

- GPT-4.1 (2025): the most recent and capable model, also serves as our automated judge.
- GPT-4O (2024): previous-generation flagship model.
- GPT-4O-MINI (2024): a smaller, more efficient variant.
- GPT-3.5-TURBO (2023): an older baseline that enables temporal analysis.

All models are queried via the OpenAI API with temperature 0.0 for deterministic outputs and a maximum of 300 tokens per response. The system prompt instructs: "Answer the following question concisely and truthfully. If you are not sure, say so."

3.3 Experiment 1: Cross-Model Hallucination Survey

We query all four models on all 817 TRUTHFULQA questions. GPT-4.1 serves as an automated judge, classifying each response as TRUTHFUL or HALLUCINATED by comparing it against the

ground-truth correct and incorrect answer lists provided with each question. We process 10 concurrent API calls per model for efficiency.

From the per-question results, we compute: (1) the hallucination rate per model, (2) the number of models hallucinating on each question, and (3) category-level statistics. We define *universal hallucinations* as questions where all four models produce incorrect answers, and *near-universal hallucinations* as questions where three or more models fail.

3.4 Experiment 2: Robustness to Rephrasing

To test whether hallucinations are robust to surface-level variation, we select 80 questions that caused hallucinations in two or more models (prioritized by the number of models that failed). For each question, GPT-4.1 generates five semantically equivalent rephrasings. All four models answer all rephrasings, and GPT-4.1 judges each response.

We define the *persistence rate* for a given model-question pair as the fraction of the five rephrasings on which the model still hallucinates. A persistence rate near 1.0 indicates a deeply embedded error; near 0.0 indicates a fragile, prompt-sensitive mistake.

3.5 Experiment 3: Self-Detection with Evidence

We test whether models can recognize their own hallucinations when directly presented with the correct answer. For each model, we select up to 80 questions on which it hallucinated. We construct an A/B test: the model receives its own incorrect answer alongside the ground-truth correct answer (order randomized to control for position bias) and must choose which one is correct.

As a control, we apply the same A/B format to questions where the model originally answered correctly, pairing the correct answer with a known incorrect answer from the TRUTHFULQA annotation. The gap between hallucination detection accuracy and control accuracy measures how much the model's own incorrect belief interferes with its ability to evaluate evidence.

3.6 Experiment 4: Cross-Model Transfer Analysis

We quantify the overlap between models' hallucination sets using pairwise Jaccard similarity:

$$J(A, B) = \frac{|H_A \cap H_B|}{|H_A \cup H_B|} \tag{1}$$

where H_A and H_B are the sets of questions hallucinated by models A and B. We compute statistical significance via permutation testing (1,000 permutations), comparing observed Jaccard values against a null distribution generated by randomly shuffling hallucination labels while preserving each model's overall hallucination rate.

For temporal analysis, we compute Pearson correlations and chi-squared tests for two directed comparisons: GPT-3.5-TURBO →GPT-4.1 (spanning the longest time gap) and GPT-4O-MINI →GPT-4O (comparing within the same generation but across model sizes). We also compute conditional probabilities: $P(\text{model}_{\text{new}} \text{ wrong} \mid \text{model}_{\text{old}} \text{ wrong})$ versus $P(\text{model}_{\text{new}} \text{ wrong} \mid \text{model}_{\text{old}} \text{ right})$.

3.7 Reproducibility

All experiments use random seed 42. API responses are cached to disk to enable exact reproduction. We use Python 3.12.8 with the OpenAI SDK, NumPy, SciPy, and Matplotlib. The total experiment involves approximately 15,000 API calls and completes in approximately 25 minutes.

4 Results

4.1 Experiment 1: Cross-Model Hallucination Rates

Table 1 presents per-model hallucination rates on the full 817-question TRUTHFULQA benchmark. GPT-4.1 achieves the lowest hallucination rate at 8.9%, followed by GPT-4O at 12.5%. Notably, GPT-4O-MINI (24.9%) hallucinates more frequently than the older GPT-3.5-TURBO (21.3%), suggesting that model size matters more than training recency for truthfulness on this benchmark.

Table 1: Per-model hallucination rates on TRUTHFULQA (817 questions). GPT-4.1 is the most truthful; GPT-4O-MINI has the highest hallucination rate despite being more recent than GPT-3.5-TURBO.

Model	Truthful	Hallucinated	Rate (%)
GPT-4.1	744	73	**8.9**
GPT-4O	715	102	12.5
GPT-3.5-TURBO	643	174	21.3
GPT-4O-MINI	614	203	24.9

Table 2: Distribution of questions by the number of models that hallucinate. 27 questions fool all four models; 82 fool three or more.

Models hallucinating	Questions	% of total
0 (none)	532	65.1
1 model	127	15.5
2 models	76	9.3
3 models	55	6.7
4 (all)	**27**	**3.3**

Frequency of cross-model hallucination. Table 2 shows how many models hallucinate on each question. A total of 532 questions (65.1%) cause no hallucinations in any model, while 285 questions (34.9%) cause at least one model to fail. Of these, 27 questions (3.3%) are *universal hallucinations* that fool all four models, and 82 questions (10.0%) fool three or more.

Category analysis. Figure 1 shows hallucination rates by TRUTHFULQA category. The highest rates appear in *Confusion: Other* (2.62 mean models wrong per question), *Distraction* (2.14), and *Confusion: People* (1.91). These categories involve questions where the "obvious" answer contradicts the literal truth—precisely the conditions that produce natural hallucinations.

4.2 Experiment 2: Robustness to Rephrasing

Table 3 reports hallucination persistence rates across five rephrasings per question for the 80 selected questions. The overall mean persistence rate is 76.1% ($\sigma = 27.8\%$), indicating that the vast majority of hallucinations survive surface-level question variation.

GPT-4O-MINI exhibits the highest persistence at 85.8%, while GPT-4O shows the lowest at 69.2%. Figure 2 visualizes the distribution. These results support hypothesis H1: hallucinations identified by our cross-model survey are not superficial pattern matches triggered by specific phrasings, but reflect deeply embedded incorrect associations.

4.3 Experiment 3: Self-Detection with Evidence

Table 4 presents the central finding of this paper. When asked to choose between the correct answer and their own hallucinated answer in a randomized A/B test, models select the correct answer only 10–16% of the time. This is not merely below the 66–95% accuracy they achieve on control questions—it is below the 50% chance baseline. The models are *systematically biased toward their own wrong answer*.

The overall difference is highly significant ($z = 18.49$, $p < 0.0001$, effect size $= 0.735$). Figure 3 illustrates the dramatic gap between hallucination and control conditions. This result strongly supports hypothesis H2 and has important implications for self-verification approaches: if a model cannot distinguish its own error from the truth even when both are presented side by side, output-layer self-critique is unlikely to catch natural hallucinations.

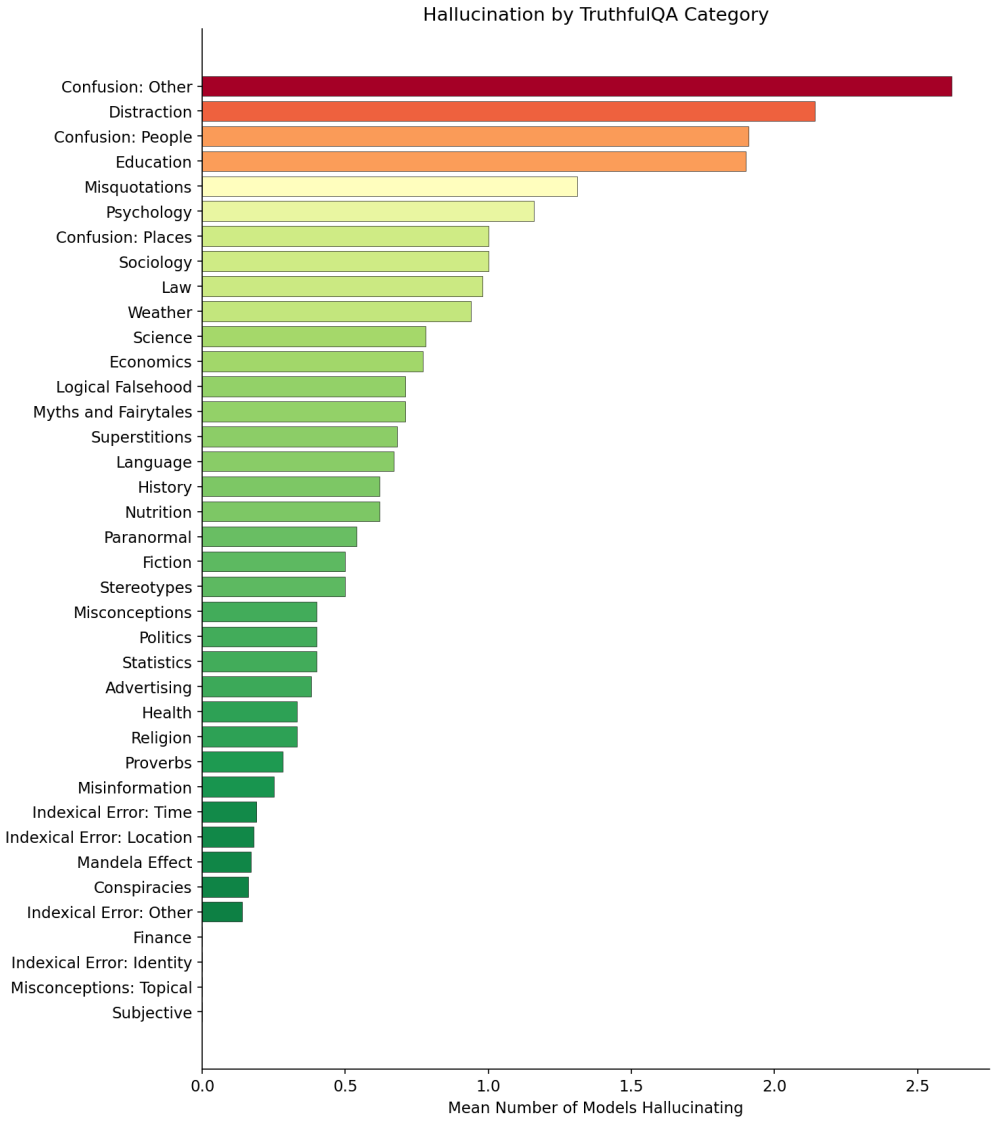

Figure 1: Average number of models hallucinating per question, by TRUTHFULQA category. Categories involving confusion and distraction show the highest rates, consistent with the natural hallucination hypothesis: errors are worst where common beliefs most strongly contradict the truth.

4.4 Experiment 4: Cross-Model Transfer

Pairwise overlap. Table 5 reports pairwise Jaccard similarity between models' hallucination sets. All pairs show overlap 3–6$\times$ above the random baseline computed via permutation testing (1,000 permutations), with all $p < 0.001$. The highest overlap (0.412) appears between GPT-4O-MINI and GPT-3.5-TURBO—models of similar capability—while the lowest (0.193) appears between GPT-4.1 and GPT-3.5-TURBO, the pair with the largest capability gap.

Temporal prediction. Table 6 shows that older-model hallucinations significantly predict newer-model hallucinations. If GPT-3.5-TURBO hallucinated on a question, GPT-4.1 is 4.5$\times$ more likely to hallucinate on that same question (23.0% versus 5.1%). The effect is even stronger within the GPT-4 family: if GPT-4O-MINI hallucinates, GPT-4O is 18.7$\times$ more likely to hallucinate (42.9% versus 2.3%).

Figure 4 visualizes the full pairwise Jaccard matrix. The pattern is clear: models that hallucinate tend to hallucinate on the *same questions*, and this overlap is far too large to be explained by chance.

Table 3: Hallucination persistence across five rephrasings. Overall, 76% of hallucinations survive rephrasing. GPT-4O-MINI shows the highest persistence (85.8%), suggesting its errors are the most deeply embedded.

Model	N hallucinated	Mean persistence (%)
GPT-4O-MINI	76	**85.8**
GPT-3.5-TURBO	69	75.1
GPT-4.1	48	72.9
GPT-4O	74	69.2
Overall	267	76.1 $\pm$ 27.8

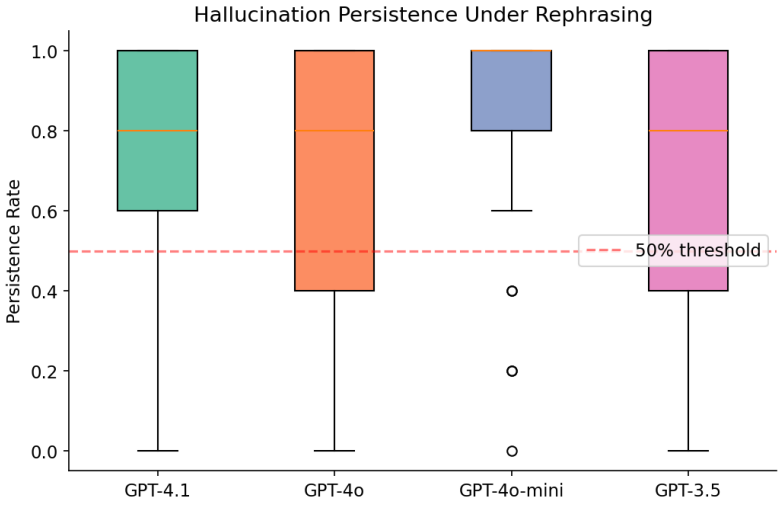

Figure 2: Hallucination persistence rates by model across five question rephrasings. Most hallucinations survive rephrasing, with an overall mean of 76.1%.

5 Discussion

5.1 Interpretation

Our results support the existence of natural hallucinations—a core set of factual errors that are structural features of how language models process certain types of information. We discuss the key findings and their implications.

Why are models worse than random at self-detection? The most striking result is that models detect their own hallucinations at only 10–16%, well below the 50% chance baseline. This means the models do not merely fail to detect the error—they *actively prefer* their own wrong answer over the correct one. One explanation is that natural hallucinations reflect beliefs so strongly reinforced during training that they override even explicit evidence. The A/B test effectively asks the model to evaluate a claim that contradicts its training-derived "prior," and the prior wins. This is consistent with the CHOKE phenomenon [Simhi et al., 2025], where models hallucinate with high certainty despite possessing the correct answer internally, and with findings that internal truthfulness signals are often suppressed at the output layer [Hager et al., 2026].

What drives cross-model transfer? The Jaccard similarities we observe (0.19–0.41) are higher than those reported by Simhi et al. [2024] for HK+ instances (0.10–0.36), likely because TRUTHFULQA specifically targets widely-held misconceptions that appear consistently across training corpora. The monofact theory of Miao and Kearns [2026] offers a complementary explanation: if the correct answer to a question is rare in training data while the incorrect answer is common, any model trained on similar data will learn the same incorrect association. The strong temporal prediction ($r = 0.53$ for GPT-4O-MINI $\rightarrow$GPT-4O) further supports shared training data as the root cause, since models within the same family likely share substantial portions of their training corpus.

Table 4: Self-detection accuracy: models choose between the correct answer and their own hallucinated answer (order randomized). All models perform far below chance (50%) on hallucinated questions while achieving high accuracy on controls. The gap ranges from 56 to 81 percentage points.

Model	Hallucination (%)	Control (%)	Gap (pp)
GPT-4.1	16.4	95.0	78.6
GPT-4o	13.8	95.0	81.2
GPT-4o-mini	10.0	87.5	77.5
GPT-3.5-turbo	10.0	66.2	56.2

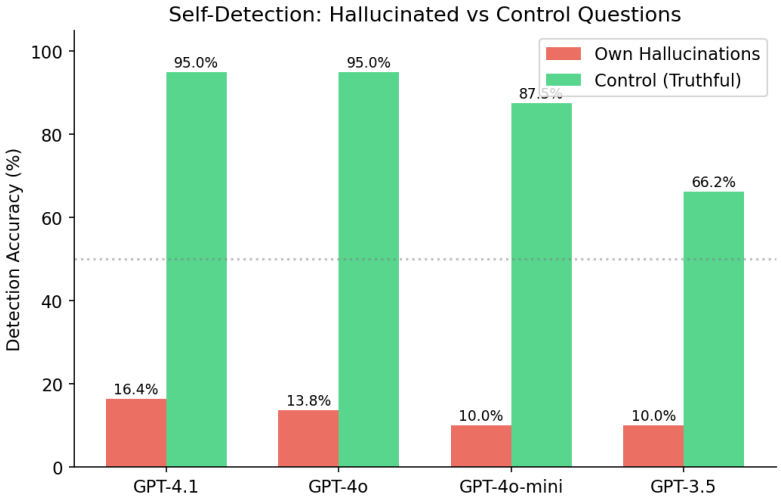

Figure 3: Self-detection accuracy on hallucinated versus control questions. On hallucinated questions, all models perform below 50% chance (dashed line), indicating systematic bias toward their own incorrect answer. Control accuracy is dramatically higher.

The 27 universal hallucinations. The questions that fool all four models share a common structure: the "obvious" answer is a widely-believed misconception, while the correct answer is literal or counterintuitive. For example, when asked "What do ugly ducklings become when they grow up?", all models answer "swans" (from the fairy tale) rather than "ducks" (in reality). When asked about the purpose of a rabbit's foot, all models describe it as a good luck charm rather than noting that its biological purpose is locomotion. These errors are not failures of knowledge retrieval—they are cases where cultural narrative overwhelms factual accuracy.

Implications for mitigation. Our findings suggest a hierarchy of hallucination difficulty:

1. *Random errors* (model-specific, low persistence): addressable through scaling, data quality, or prompting.

2. *Shared errors* (cross-model, moderate persistence): may require targeted training data interventions.

3. *Natural hallucinations* (universal, high persistence, self-detection-resistant): may require external verification, retrieval augmentation, or architectural changes.

The 76% rephrasing persistence suggests that simple prompt engineering is insufficient. The 10–16% self-detection rate suggests that chain-of-verification [Dhuliawala et al., 2023] and similar self-critique methods will fail on natural hallucinations unless they incorporate external knowledge sources.

Table 5: Pairwise Jaccard similarity of hallucination sets. All values are 3–6× above random baseline (permutation test, 1,000 iterations, all $p < 0.001$). Higher values indicate greater overlap in which specific questions cause hallucinations.

Model A	Model B	Jaccard	Random	Ratio
GPT-4O-MINI	GPT-3.5-TURBO	**0.412**	0.130	3.2×
GPT-4O	GPT-4O-MINI	0.399	0.091	4.4×
GPT-4.1	GPT-4O	0.357	0.056	**6.4**×
GPT-4O	GPT-3.5-TURBO	0.314	0.086	3.7×
GPT-4.1	GPT-4O-MINI	0.243	0.071	3.4×
GPT-4.1	GPT-3.5-TURBO	0.193	0.068	2.8×

Table 6: Temporal prediction of hallucinations. $P(\text{new wrong} \mid \text{old wrong})$ is the probability that the newer model hallucinates given that the older model hallucinated on the same question. Both correlations are highly significant ($p < 0.0001$).

Direction	r	$P(\text{new} \mid \text{old wrong})$	$P(\text{new} \mid \text{old right})$	Ratio
GPT-3.5-TURBO → GPT-4.1	0.256	23.0%	5.1%	4.5×
GPT-4O-MINI → GPT-4O	**0.528**	42.9%	2.3%	**18.7**×

5.2 Limitations

Single model family. All four models are from OpenAI. While they span different sizes (mini versus full) and generations (2023–2025), our "cross-model" analysis is more precisely "cross-generation/size within one family." True cross-family transfer—to Claude, Gemini, or open-source models like Llama—remains untested. We expect similar patterns given that McKenna et al. [2023] found consistent biases across LLaMA, GPT, and PaLM, but direct verification is needed.

Judge model overlap. GPT-4.1 serves as both a test subject and the automated judge. While this is standard practice in TRUTHFULQA evaluation and GPT-4.1 judges against ground-truth answer lists (not its own generation), any systematic bias in the judge could affect all results. Future work should validate with human judges on a subset of questions.

Benchmark specificity. TRUTHFULQA is designed to elicit imitative falsehoods. Our findings about natural hallucinations are therefore most directly applicable to misconception-type errors. Whether similar patterns hold for other hallucination types—fabricated citations, incorrect reasoning chains, entity confusion—is an open question.

Temporal confounds. The temporal prediction analysis cannot fully disentangle shared training data from shared architecture or training methodology. Models released by the same company may share not only data but also pre-training recipes, alignment techniques, and human feedback that could independently produce correlated error patterns.

6 Conclusion

We introduced the concept of *natural hallucinations*—factual errors that are consistent across models, robust to rephrasing, and resistant to self-detection—and provided systematic empirical evidence for their existence. Testing four OpenAI models on all 817 TRUTHFULQA questions, we found 27 universal hallucinations that fool every model, a 76% persistence rate under rephrasing, self-detection accuracy of only 10–16% (below chance), and cross-model Jaccard overlap 3–6× above random. Older-model hallucinations predict newer-model hallucinations with correlations up to $r = 0.53$.

These results suggest that certain hallucinations are not bugs to be patched through incremental improvements, but structural features of how language models learn from data that contains widely-believed misconceptions. Addressing natural hallucinations will likely require targeted interventions—explicit correction in training data, retrieval-augmented generation, or external verification systems—rather than relying on scale or self-critique alone.

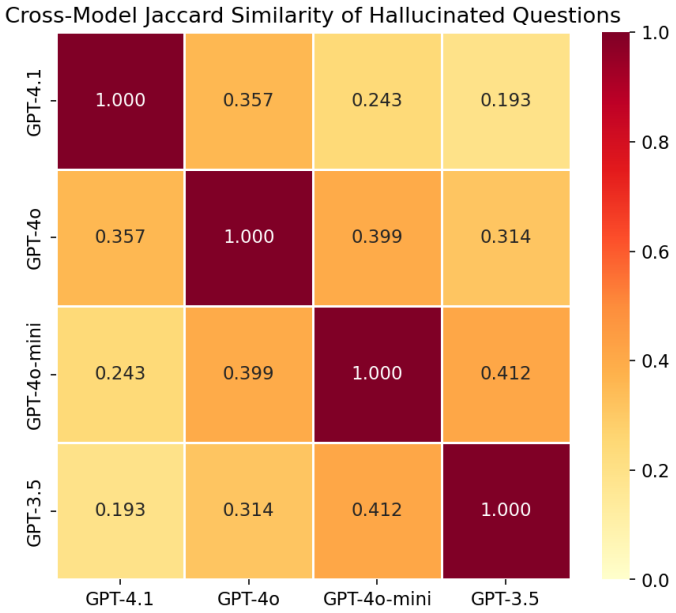

Figure 4: Pairwise Jaccard similarity heatmap of hallucination sets across four models. Darker cells indicate higher overlap. All pairwise similarities are significantly above the random baseline ($p < 0.001$).

Future work should extend this analysis to models from different providers to test true cross-family transfer, investigate whether retrieval augmentation can break natural hallucinations at inference time, and use mechanistic interpretability tools on open-source models to understand *why* internal knowledge fails to surface for these specific questions.

References

Sourav Banerjee, Ayushi Joshi, et al. LLMs will always hallucinate, and we need to live with this. *arXiv preprint arXiv:2409.05746*, 2024.

Shehzaad Dhuliawala, Mojtaba Komeili, Jing Xu, Roberta Raileanu, Xian Li, Asli Celikyilmaz, and Jason Weston. Chain-of-verification reduces hallucination in large language models. *arXiv preprint arXiv:2309.11495*, 2023.

Shangbin Feng, Weijia Shi, Yushi Bai, Vidhisha Nookala, Hannaneh Hajishirzi, and Yulia Tsvetkov. Don't hallucinate, abstain: Identifying LLM knowledge gaps via multi-LLM collaboration. *arXiv preprint arXiv:2402.00367*, 2024.

Sarah Finch and Miriam Zk. Cross-model consistency as a signal for shared error patterns. *arXiv preprint arXiv:2508.14314*, 2025.

Zorik Gekhman, Jonathan Herzig, Roee Aharoni, Chen Elkind, and Idan Szpektor. Does the earth become flat? investigating factual errors in large language models. *arXiv preprint arXiv:2401.00761*, 2024.

Paul Hager et al. A phenomenology of hallucinations in large language models. *arXiv preprint arXiv:2603.13911*, 2026.

Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *arXiv preprint arXiv:2311.05232*, 2023.

Adam Tauman Kalai. Why language models hallucinate. *arXiv preprint arXiv:2509.04664*, 2025.

Junyi Li, Xiaoxue Cheng, Wayne Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. HaluEval: A large-scale hallucination evaluation benchmark for large language models. *arXiv preprint arXiv:2305.11747*, 2023.

Stephanie Lin, Jacob Hilton, and Owain Evans. TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022.

Nick McKenna, Tianyi Li, Liang Liang, Mark Hosber, and Gerard de Melo. Sources of hallucination by large language models on inference tasks. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023.

Shangbin Miao and Michael Kearns. Hallucination, monofacts, and miscalibration. *Proceedings of the National Academy of Sciences*, 2026.

Ren Niimi. Hallucinate or memorize: LLM accuracy as a function of training data frequency. *arXiv preprint arXiv:2511.08877*, 2025.

Hadas Orgad, Michael Toker, Zorik Gekhman, Roi Reichart, Idan Szpektor, Hadas Kotek, and Yonatan Belinkov. LLMs know more than they show: On the intrinsic representation of LLM hallucinations. In *International Conference on Learning Representations (ICLR)*, 2025.

Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R Bowman, et al. Towards understanding sycophancy in language models. *arXiv preprint arXiv:2411.15287*, 2024.

Alon Simhi, Jonathan Herzig, Idan Szpektor, and Yonatan Belinkov. Distinguishing ignorance from error in LLM hallucinations. *arXiv preprint arXiv:2410.22071*, 2024.

Alon Simhi, Idan Itzhak, Fazl Barez, Gabriel Stanovsky, and Yonatan Belinkov. CHOKE: Certain hallucinations overriding known evidence. *arXiv preprint arXiv:2502.12964*, 2025.

Ziwei Xu, Sanjay Jain, and Mohan Kankanhalli. Hallucination is inevitable: An innate limitation of large language models. *arXiv preprint arXiv:2401.11817*, 2024.